

International Transfers or Differentiated Carbon Prices?

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Abstract

Recent proposals for international carbon pricing differentiate national carbon prices by income. The usual justification is that cross-border transfers are politically infeasible. But differentiated prices are themselves equivalent to transfers. We show that any schedule of national prices is distributionally equivalent to a uniform price combined with a certain allocation of tradable emission rights, equal at the first order to each country's emissions under autarky. In a dynamic setting, the equivalent rights are weighted by the discounted carbon price, and under Hotelling pricing they reduce to a carbon budget. Because abatement costs are convex, this allocation leaves every country weakly better off, so the uniform price with equivalent rights Pareto-dominates autarky. We quantify the equivalence in NICE2020, an integrated assessment model with 179 countries. Replacing the schedules of [Parry et al. \(2021\)](#), [Wolfram et al. \(2025\)](#) or [Banerjee et al. \(2025\)](#) with a uniform price and equivalent rights would cut the coalition's emissions by about 5% while leaving every member as well off. Alternatively, holding emissions fixed, it would raise every member's consumption. We also review the departures from standard assumptions that could justify price differentiation and argue that they are not compelling.

Keywords: carbon pricing; international climate agreements; emissions trading; burden sharing; differentiated carbon prices; international transfers; integrated assessment.

JEL: Q54; Q58; H23; F53; D61.

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1 Introduction

International climate policy is stuck on a distributional question: who should obtain the rights to emit? The Paris Agreement endorses common but differentiated responsibilities and respective capabilities, and it commits developed countries to financing climate action in developing ones. It does not translate these principles into numbers, and national contributions remain far short of its temperature goal (UNEP 2025).

A prominent recent answer is to coordinate carbon prices while differentiating them by income. The IMF proposes an international price floor of \$75, \$50 and \$25/tCO₂ for high-, upper middle- and lower-income countries (Parry et al. 2021). Wolfram et al. (2025) propose a climate coalition built on the same

tiers. [Banerjee et al. \(2025\)](#) propose a “grand bargain” in which low-, and middle-income countries adopt prices of \$10 to \$50/t in exchange for the compensation of their climate damages. In all three proposals, carbon revenues are retained domestically. Differentiation is meant to deliver equity without the cross-border transfers that are deemed politically infeasible ([Parry et al. 2021](#)).

Textbook theory prescribes something else: a uniform price for efficiency, and lump-sum transfers for equity ([Aldy & Stavins 2012](#)). Differentiated prices are a second-best substitute for transfers, and the case for income-based differentiation therefore rests on the premise that transfers are infeasible. This paper argues that the premise does not rest on solid ground. Survey evidence already challenges it: majorities accept an international carbon price financing North–South transfers in all 23 surveyed countries ([Fabre 2025](#); [Fabre et al. 2025](#)). Furthermore, a uniform price can achieve efficiency while remaining distributionally equivalent to differentiated prices. We make the claim precise in three results.

First, any schedule of differentiated autarky prices is distributionally equivalent to a uniform price combined with a specific allocation of tradable emission rights. To first order, this allocation grants each country the emissions it has under autarky. A country allowed a price below the uniform price therefore receives more rights than it emits at the uniform price, and sells the difference. Conversely, a positive price gap is equivalent to less emission rights than in autarky. In a static setting, we derive a second-order formula in terms of the price gap and the price elasticity of emissions (Proposition 1).

Second, granting each country its emissions under the schedule leaves every country weakly better off under the uniform price than under autarky (Proposition 2). This is the familiar gains-from-trade logic of emissions markets ([Montgomery 1972](#)) and the international analogue of production efficiency ([Diamond & Mirrlees 1971](#)). Its implication for the policy debate is less familiar: any price-differentiation agreement can be replaced by a uniform price with an allocation of rights such that no country loses. The resulting surplus can take the form of reduced emissions or of increased consumption.

Third, in a dynamic setting the equivalence holds in present value, with each year’s rights weighted by the discounted carbon price (Proposition 3). When the carbon price follows a Hotelling path, the equivalent allocation is a carbon budget: each country’s cumulative emissions under autarky.

We quantify the equivalence in NICE2020 ([Young-Brun et al. 2025](#)), an integrated assessment model with 179 countries, in two exercises. First, for eight diverse economies, we compare welfare under a differentiated autarky price and under a global uniform price with a given allocation, holding global emissions on a path that exhausts a carbon budget of 1000 GtCO₂ over 2025–2100. For each autarky price and each allocation, we compute the welfare difference and trace the indifference curve between the two regimes. The curves illustrate that the equivalence runs only one way: many allocations of rights, including an equal per

capita allocation, cannot be replicated by any schedule of non-negative prices, because a zero price is not enough to compensate low emitters. For example, Nigeria would prefer a uniform price to any non-negative autarky price, including a zero price, as long as its emission rights per capita are at least 28% of the global average.

Second, we compute the equivalent rights for prominent differentiated prices proposals. Granting all members their equivalent rights cuts coalition emissions by 4.8% (Banerjee et al. 2025), 4.8% (Wolfram et al. 2025) and 13.4% (Equal Right 2023), while leaving every member’s consumption as high as under the schedule.

The remainder of the paper is organised as follows. Section 2 reviews the rationales for differentiated carbon prices in a world that departs from standard assumptions. Section 3 derives the equivalence between differentiated prices and emission rights, in closed form in the static case and under additional assumptions in the dynamic case. Section 4 uses NICE2020 to trace, for eight major economies, the rights that leave each one indifferent to any given autarky price. Section 5 computes the rights equivalent to prominent differentiated-price proposals, and the emissions cut or consumption gain obtained by replacing them with a uniform price. Section 6 concludes.

2 Why differentiate carbon prices?

With complete markets and no other distortion, efficiency requires a uniform carbon price, and equity should be addressed via lump-sum transfers (Sandmo 2007). Four departures from standard assumptions have been invoked to justify differentiating carbon prices across countries or sectors: divergent discount rates, pre-existing distortive taxes, market power in trade, and constraints on cross-country transfers. We review them in turn.

First, Anthoff et al. (2026) show that equalising carbon prices across countries is inefficient in case of ‘cross-country differences in discount rates (or interest rates)’, which arise from inefficiencies in the allocation of capital. By Hotelling’s argument, each country’s carbon price should grow at its own discount rate, so prices cannot be equal at all dates.¹ The authors stress, however, that their result cannot say whether trade in emission permits from a given allocation would raise or lower welfare, as this depends on how permit trade interacts with the capital-market frictions, which no model has yet specified.

Second, pre-existing tax distortions can be mitigated by differentiating carbon prices across sectors or countries (Babiker et al. 2004; Boeters 2014). For example, if aviation remains under-taxed, a higher carbon price on aviation than on other

¹Differentiated prices may then be optimal in a second-best world, but the first best is in principle attainable by correcting the capital-market imperfections themselves.

sectors improves welfare. For this reason (applied to countries), Babiker et al. (2004) show that a uniform price may not always be beneficial to the countries that take part in it. However, D'autume et al. (2016) show that a uniform price remains optimal if the other tax (such as a gasoline tax) addresses a local externality or if public goods have to be funded by distortive taxation.

Third, large countries have market power in world markets, which gives them a terms-of-trade motive to differentiate (Babiker 2005; Boeters 2014). Take China, a large oil importer and manufacturing exporter. By taxing oil more heavily than other goods, China would depress world oil demand and hence the oil price, improving its terms of trade. Similarly, it has an interest in setting a higher carbon price on manufactured goods, to raise the price of its exports (to the extent the carbon price paid by importers exceeds the loss of exports due to higher prices).

Fourth, and most commonly, differentiated prices are justified by the assertion that international transfers are politically infeasible (Parry et al. 2021; Young-Brun et al. 2025). Bauer et al. (2020) describe an “efficiency–sovereignty” trade-off in climate policy: either carbon prices are differentiated and efficiency is lost, or a uniform price with transfers is applied, hampering national sovereignty. This reasoning assumes that international transfers constitute a stronger infringement on sovereignty than coordinated price differentiation (see also Bekkers & Caricola 2022). Because transfers are deemed infeasible for they infringe sovereignty, differentiated prices are viewed as a necessary second-best solution.

The first three rationales have an efficiency basis and do not rest on the absence of transfers. They also call for a differentiation that bears no systematic relation to income: by discount rate, by sector or by market power. This paper addresses the fourth rationale, the one invoked by differentiation proposals. We show that it cannot rest on the distributional implications of transfers, since differentiation entails implicit costs or benefits of the same size as the transfers that a uniform price with equivalent rights would involve.

Related literature. Our analysis builds on the literature that evaluates specific differentiated price floors. In a global general equilibrium model, Chateau et al. (2024) find that the IMF’s international carbon price floor (Parry et al. 2021) achieves higher welfare than a uniform price *without transfers* attaining the same emission reductions (at \$56/t), since it shifts part of the mitigation effort from low- to high-income countries. They also find that it protects the competitiveness of high-income countries as well as a border carbon adjustment would. He et al. (2024) show that, relative to national pledges, the IMF price floors bind on China and India but not on high-income countries, which makes the arrangement unattractive to developing countries.

Le Moigne et al. (2026) find that income-differentiated floors do not necessarily yield fairer outcomes. Compared with a uniform tax accompanied by

the North–South transfers that equalise costs across countries (\$107–169 billion a year), all variants of the IMF floor entail larger global real-income losses, and most of them require larger transfers to reach the same fairness. We provide the first mapping from such schedules to equivalent allocations of emission rights.

Our results also speak to the literature on negotiating a common carbon price or a global cap-and-trade with revenue sharing (Cramton et al. 2017; Fabre 2024; Gollier & Tirole 2015; Grubb 1990; Nordhaus 2015; Weitzman 2014). The equivalence formula gives climate negotiators a closed-form exchange rate between price concessions and emission rights.

3 The equivalence between differentiated prices and emission rights

3.1 Static setting

Consider countries $i = 1, \dots, I$ with populations N_i , in a single period. Lower-case variables are per capita. Country i emits e_i at abatement cost $a_i(e_i)$, which is decreasing and convex in e_i . Facing a carbon price p , firms and households emit $e_i(p)$, optimising by equating the marginal abatement cost to (minus) the price: $a'_i(e_i(p)) = -p$. Consumption is

$$c_i = y_i - a_i(e_i) + t_i, \quad (1)$$

where y_i is potential output and t_i a net international transfer. Welfare is increasing in c_i and decreasing in global emissions through climate damages.

We compare two regimes that deliver the same global emissions $E = \sum_j N_j e_j$, whose world average per capita is $\bar{e} \equiv E/N$ (where N is the world population). Damages are then identical across regimes, and a country is better off in one regime if and only if its consumption is higher.

Differentiated prices (autarky). Country i applies a domestic price p_i , keeps its revenue and receives no transfer. It emits $e_i^A \equiv e_i(p_i)$, and $\sum_i N_i e_i^A = E$.

Uniform price. The same countries apply a uniform price p^* and trade emission rights under a cap equal to E . Country i receives rights r_i per capita. At the market-clearing price, $\sum_i N_i e_i(p^*) = E$, and i emits $e_i^* \equiv e_i(p^*)$. Its net transfer is the value of its sales minus its purchases of rights, $t_i = (r_i - e_i^*) p^*$.

Throughout, a star denotes the uniform-price regime, a superscript A the autarky regime and a bar a world average. We write $\pi_i \equiv p_i/p^*$ for country i 's

relative price, and $\varepsilon_i \equiv -\frac{p^*}{e_i^*} e_i'(p^*)$ for the price elasticity of its emissions at the uniform price. Define

$$b_i \equiv a_i(e_i^A) - a_i(e_i^*) - a_i'(e_i^*) (e_i^A - e_i^*) \geq 0. \quad (2)$$

b_i is the deadweight loss of the price wedge $p_i - p^*$, i.e. the Harberger triangle of country i : the area enclosed by the country's marginal abatement cost curve, the horizontal line at p^* and the vertical line at e_i^A . Formally, b_i is the Bregman divergence of a_i between e_i^A and e_i^* . It is nonnegative by convexity, and strictly positive if a_i is strictly convex and $p_i \neq p^*$.

Denote r_i^* the *equivalent rights* that leave a country i exactly as well off under the uniform-price regime as under the autarky regime with a price p_i .

Proposition 1 (Equivalent rights). *Holding total emissions at E and the carbon price at p^* , we have*

$$r_i^* = e_i^A - \frac{b_i}{p^*}. \quad (3)$$

To first order in the price gap,

$$r_i^* \approx e_i^A \approx e_i^* \left[1 + \varepsilon_i(1 - \pi_i) \right]. \quad (4)$$

To second order, with ε_i evaluated at p^* ,

$$r_i^* \approx e_i^A - \frac{(e_i^A - e_i^*)^2}{2 \varepsilon_i e_i^*} \approx e_i^* \left[1 + \varepsilon_i(1 - \pi_i) - \frac{1}{2} \varepsilon_i(1 - \pi_i)^2 \right]. \quad (5)$$

Proof. Equating $y_i - a_i(e_i^*) + (r_i - e_i^*)p^*$ with $y_i - a_i(e_i^A)$ gives $r_i = e_i^* + [a_i(e_i^*) - a_i(e_i^A)]/p^*$. Substituting $a_i'(e_i^*) = -p^*$ into (2) yields (3). Equation (5) follows from $e_i^A - e_i^* \approx \varepsilon_i e_i^*(1 - \pi_i)$. For (5), note that $a_i''(e_i^*) = -1/e_i'(p^*) = p^*/(\varepsilon_i e_i^*)$, so that $b_i \approx \frac{1}{2} p^* (e_i^A - e_i^*)^2 / (\varepsilon_i e_i^*)$. $\square$

Proposition 1 shows that a differentiated price is a transfer paid in kind. Relative to the uniform price, a country allowed a lower price ($\pi_i < 1$) keeps emissions $e_i^A > e_i^*$. To first order, the uniform-price regime replicates its situation by granting it exactly those emissions as rights, part of which it then sells, receiving $(e_i^A - e_i^*)p^*$. Conversely, a country asked for a higher price ($\pi_i > 1$) is in the position of a country that buys rights. The size of the implicit transfer grows with the price gap, the country's emissions and the elasticity of those emissions. The same formula, read in reverse, converts any allocation of rights into an equivalent autarky price.

It is convenient to express an allocation as a multiple of the world average, $r_i = \rho_i \bar{e}$, so that $\rho_i = 1$ is an equal per capita allocation. We denote by $\rho_i^* \equiv r_i^* / \bar{e}$ the equivalent ratio, and by $\hat{\rho}_i$ its first-order approximation:

$$\hat{\rho}_i \equiv \frac{e_i^A}{\bar{e}}, \quad (6)$$

country i 's per capita emissions under the differentiated schedule relative to the world average. By (3), $\rho_i^* = \hat{\rho}_i - b_i / (p^* \bar{e}) \leq \hat{\rho}_i$: the first-order prediction errs on the generous side. A differentiated schedule is thus, to first order, a grandfathering rule, applied to the emissions that the schedule itself induces.

Proposition 2 (Dominance of the uniform price). *Allocate rights $r_i = e_i^A$ to every country. This allocation (i) is feasible at the cap of the autarky regime, $\sum_i N_i r_i = E$; (ii) leaves every country weakly better off than under the autarky regime, strictly so if a_i is strictly convex and $p_i \neq p^*$; and (iii) leaves a surplus of $\sum_i N_i b_i \geq 0$. Expressed in emissions, the rights can be cut to the equivalent allocation $(r_i^*)_i$, with no country worse off than under differentiated prices.*

Proof. Part (i) holds by construction. For part (ii), the consumption gain of country i is $[a_i(e_i^A) - a_i(e_i^*)] + (e_i^A - e_i^*)p^* = b_i \geq 0$. For part (iii), sum these gains. $\square$

A country that is allowed a lower price saves on abatement costs. Under the uniform price, the same country can abate up to p^* and sell the freed rights at p^* , which is always at least as good a deal. By Proposition 1, the surplus is of second order in the price gaps, whereas the implicit transfers are of first order. The efficiency argument for a uniform price is therefore robust but quantitatively modest when prices are not very dispersed, whereas the distributional stakes of the allocation are large.

Proposition 2 holds total emissions at $E = \sum_i N_i e_i^A$ and the uniform price at p^* : turning the surplus into a lower cap raises the price, so that in general equilibrium the equivalent rights are slightly greater than $e_i^A - b_i / p^*$. Therefore, the attainable emissions cut differs from $\sum_i N_i b_i / p^*$, which is why we solve for it numerically in Section 5.

Proposition 2 is a first-best statement. It abstracts from the second-best rationales for differentiation reviewed in Section 2. None of these requires the absence of transfers. What it questions is the argument that differentiation is needed because transfers would be infeasible: differentiation involves the same distributive outcome as well-chosen transfers; it only makes them implicit.

3.2 Dynamic setting

With several periods t , prices p_{it} , rights and emissions become trajectories, populations N_{it} change over time, and β_t denotes the discount factor. Under total

utilitarianism, country i 's objective is $\sum_t \beta_t N_{it} u(c_{it})$. It is therefore convenient to state the dynamic results in totals rather than per capita: $R_{it} \equiv N_{it} r_{it}$ and $E_{it} \equiv N_{it} e_{it}$ are country i 's rights and emissions in period t , E_{it}^A its emissions under the autarky regime, and $E_t = \sum_i E_{it}$ world emissions, so that the population weights are carried by the quantities themselves.

Proposition 3 (Dynamic equivalence). *Suppose abatement costs are separable over time. To first order, country i is indifferent between the autarky regime and the uniform price path (p_t^*) with rights (R_{it}) if and only if*

$$\sum_t \beta_t p_t^* (R_{it} - E_{it}^A) = 0. \quad (7)$$

If rights are a constant multiple ρ_i of an equal per capita share, $r_{it} = \rho_i \bar{e}_t$, the equivalent ratio is, to first order,

$$\hat{\rho}_i = \frac{\sum_t \beta_t p_t^* N_{it} e_{it}^A}{\sum_t \beta_t p_t^* N_{it} \bar{e}_t} : \quad (8)$$

the country's emissions under autarky over its population share of world emissions, each period weighted by the discounted price.

Corollary 1 (Hotelling case). *If the discounted price $\beta_t p_t^*$ is constant over time, the dynamic equivalence (7) only constrains cumulative rights $R_i \equiv \sum_t R_{it}$, which must equal the country's carbon budget under the autarky regime, and which admits a first-order approximation in function of the price ratio π_i analogous to the static case if differentiated prices are homothetic, $p_{it} = \pi_i p_t^*$:*

$$R_i = E_i^A \approx E_i^* \left[1 + \varepsilon_i (1 - \pi_i) \right], \quad (9)$$

where $E_i^ \equiv \sum_t E_{it}^*$ (resp. $E_i^A \equiv \sum_t E_{it}^A$) denotes i 's cumulative emissions under the uniform price (resp. autarky), and $\varepsilon_i \equiv \sum_t \varepsilon_{it} E_{it}^* / E_i^*$ denotes the average over time, weighted by these emissions, of its price elasticities ε_{it} at p_t^* . Consequently,*

$$\hat{\rho}_i = \frac{E_i^A}{\sum_t \frac{N_{it}}{N_t} E_t}, \quad (10)$$

that is the ratio of country i 's share of the cumulative emissions under autarky to its equal per capita share of the world carbon budget.

Proof. Apply the first-order argument of Proposition 1 period by period. In period t the consumption gain per capita is $p_t^*(r_{it} - e_{it}^A)$ to first order; under total utilitarianism it enters the country's objective multiplied by its population, which turns per capita quantities into totals, $N_{it}(r_{it} - e_{it}^A) = R_{it} - E_{it}^A$, and it is

discounted by β_t . Summing over t and setting the total to zero gives (7). Substituting $R_{it} = \rho_i(N_{it}/N_t)E_t$ into (7) gives (8), and $\beta_t p_t^*$ cancels from it when it is constant, leaving (10). Likewise, a constant $\beta_t p_t^*$ factors out of (7), which then reads $\sum_t (R_{it} - E_{it}^A) = 0$. If $p_{it} = \pi_i p_t^*$, then (4) applied in each period gives $E_{it}^A \approx E_{it}^*[1 + \varepsilon_{it}(1 - \pi_i)]$, and summing over t gives (9). $\square$

Equation (7) shows that timing matters. A tonne of rights is worth more in a period of high (discounted) prices, so the allocation depends on the whole price path, and a country's cumulative carbon budget does not by itself determine its welfare. Countries whose emissions fall quickly under autarky need fewer rights than their cumulative share suggests, to the extent that the discounted price is higher in the future when their emissions are low. Only under a Hotelling price path, the natural benchmark with a fixed carbon budget and efficient capital markets, does the price weighting drop out, and (9) then states that a differentiated price is worth to a country the domestic carbon budget it induces.

4 Equivalent rights by country

4.1 Model

We use NICE2020 (Young-Brun et al. 2025), a recursive-dynamic integrated assessment model in the RICE tradition. It represents 179 countries, each disaggregated into income deciles with specific carbon footprints and climate damages, and is coupled to the FaIR climate model. Damages are country-specific: the damage function is applied to local temperatures. Abatement costs are not: all countries share the same cost function of the abatement rate and differ only through their baseline emissions. A given price therefore induces the same proportional abatement everywhere, so that the price elasticity ε_i of Proposition 1 is common to all countries at a given price, and cross-country differences in equivalent rights stem from differences in emissions and in autarky prices.

We present every result under two variants, which differ in how a country's situation is measured and in how carbon revenue is shared within it. In the *consumption* variant, our benchmark, we compare a country's situation in two regimes by the net present value over 2025–2100, discounted at 3% a year, of its total consumption, i.e. consumption per capita times population, summed over the country's income deciles. Domestic carbon revenues are recycled within each country in proportion to c^η , the inverse of marginal utility (Negishi weights, with $\eta = 1.5$). This rule roughly preserves within-country inequality, and the criterion ignores it, so that the comparisons isolate between-country effects. In the *welfare* variant, the criterion is the net present value of population-weighted equally-distributed-equivalent (EDE) consumption: the level of consumption which, equally

distributed across deciles, would yield the same utilitarian welfare as the actual distribution, with inequality aversion $\eta = 1.5$. Revenues are returned as an equal per capita dividend within each country, which makes the domestic incidence of a carbon price progressive. This variant thus accounts for the within-country effects of carbon pricing that the first one abstracts from. In both, population weighting is what the total-utilitarian framework of Section 3 requires: a year in which a country is more populous counts for more. We show the consumption variant in the main text, report the main figures of the welfare variant alongside, and give its full results in Online Appendix D. International transfers enter national income and therefore affect investment and subsequent growth. Online Appendix A gives further details.

4.2 Design

The benchmark is an exponential uniform price path p_t^* , ramped up linearly from zero in 2025 over ten years and growing at a constant rate from 2036, which exhausts a carbon budget of 1000 GtCO₂ over 2025–2100. Among such paths, we take the one that maximises discounted world welfare, damages included, over 2030–2300.² The ramp-up reaches \$146/t in 2035, after which the price grows at 1.6% a year, reaching \$182/t in 2050 and \$343/t in 2090. World emissions fall from 38 GtCO₂ in 2025 to 28 in 2030, 21 in 2035 and 5 in 2090, and warming reaches 1.84°C in 2100. The price grows more slowly than the 3% discount rate, so that the discounted price falls over time: with damages in the objective, a tonne abated early is worth more than the carbon budget alone would imply. For a given country i , we compare two scenarios with the same global emission path.

- *Autarky*: country i applies $\pi_i p_t^*$ and recycles its revenue domestically. The rest of the world applies a uniform price that we solve for year by year so that global emissions stay on the path of the uniform-price run.
- *Uniform price*: all countries apply p_t^* . Country i receives rights per capita ρ_i times its population share of global emissions in the same year, $r_{it} = \rho_i \bar{e}_t$, and the remaining rights go to the rest of the world in proportion to each country's emissions.

A grid over (π_i, ρ_i) gives the net gain in consumption (in the consumption variant) or welfare (in the welfare variant) between the two scenarios, and hence country i 's indifference curve. We run the grid for eight diverse (sets of) countries.

²Welfare is discounted at a pure rate of time preference of 0.3%, which, with $\eta = 1.5$ and the model's 1.8% annual growth of consumption per capita, corresponds to the 3% discount rate on consumption used throughout.

4.3 Results

Figure 1 maps each country’s consumption gain in the uniform-price regime relative to autarky over the (π_i, ρ_i) plane, in the consumption variant; Figure A1 in the Online Appendix is its welfare-variant analogue. Its indifference curve gives, for each autarky price factor, the smallest rights allocation at which the country prefers the uniform price. The first panel repeats the United States with the y-axis on a linear rather than a logarithmic scale: the curve is then close to a straight line, as equation (4) implies, since rights are valued at the uniform price and the allocation that compensates a price gap is proportional to it.

Current emissions poorly predict equivalent rights. The intercept of each curve at $\pi_i = 1$, denoted ρ_1 , is consistent with the theory: when the autarky price equals the uniform price, the equivalent rights closely match the country’s cumulative emissions.³ It is worth noting that the equivalent rights account for future projected emissions while a 2025 snapshot does not. In 2025, China emits 1.7 times the world average per capita, yet its equivalent allocation is below an equal per capita share ($\rho_1 = 0.95$), because its emissions are modelled to fall steeply. Conversely, India emits 0.4 times the world average but is equivalent to 0.91 shares, because its emissions hold up. The EU27, at 1.4 times the world average, is equivalent to only 0.37 shares (Table 1). The welfare variant gives nearly the same intercepts, within 0.03 of these for every country (Online Appendix D): at $\pi_i = 1$ the two regimes differ only by the allocation, whose within-country incidence is the same in both.

Fair allocations cannot be replicated by differentiated prices. While every price schedule has an equivalent allocation, not every allocation has an equivalent non-negative price. At a zero price, a country cannot emit more than its unabated emissions, so the most that differentiation can offer it is the allocation $\rho_i(0)$ at which its curve meets the vertical axis. For low emitters, that bound lies far below an equal per capita share: 0.28 for Nigeria and 0.05 for the DRC. In the welfare variant, the bound is tighter still: with an equal per capita dividend, a domestic carbon price also redistributes within the country, which the welfare criterion values, so the equivalent allocation is highest near $\pi_i = 1$ rather than at a zero price, and never exceeds 0.19 for Nigeria and 0.04 for the DRC (Online Appendix D). Burden-sharing principles that give such countries an equal per capita share or more — as historical responsibility or capacity to pay would — can only be achieved through a cap-and-trade or through monetary transfers.

³The residual gaps are due to the homotheticity constraint on rights ($r_{it} = \rho_i \bar{e}$), which creates temporary price gaps that in turn trigger channels ignored by the first-order formula: the convexity of abatement costs, the effect of transfers on capital accumulation, and the within-country

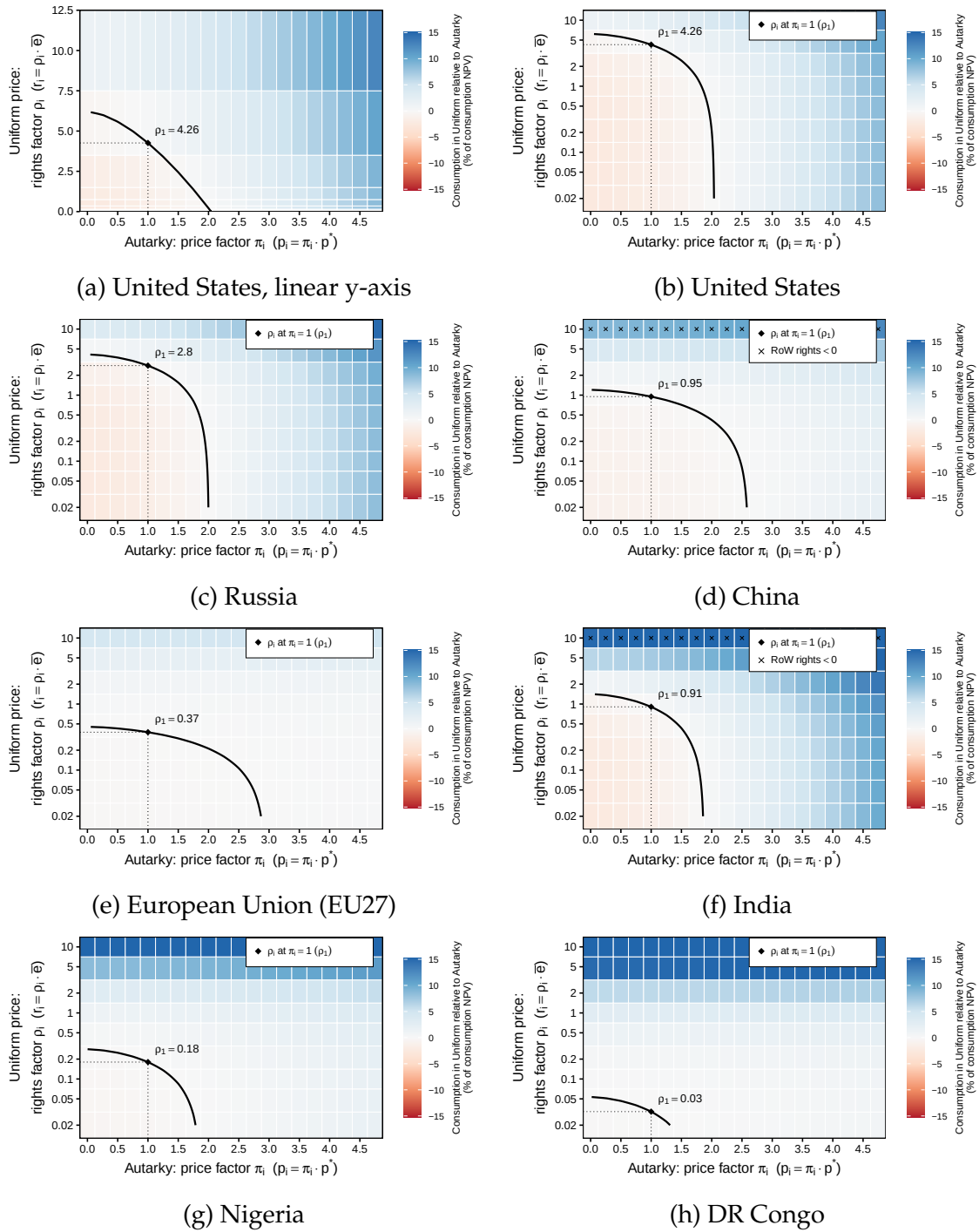


Figure 1: Consumption gain of the uniform-price regime over autarky

Note: The horizontal axis is country i 's autarky price as a multiple π_i of the benchmark price p_i^* , from 0 (no domestic carbon price) to 4.75, in steps of 0.25. The vertical axis is its allocation ρ_i under the uniform price, as a multiple of world average emissions per capita: on a log scale in every panel but the first, which repeats the United States on a linear scale to show that the indifference curve is close to a straight line. The axes are otherwise the same in every panel. Colours give the difference in the NPV of consumption over 2025–2100, in percent, on a common scale across countries clipped at the 95th percentile of the absolute difference (± 15.1); blue cells favour the uniform price. The black line is the indifference curve and ρ_1 its value at $\pi_i = 1$. Global emissions are the same in both regimes at every point of the grid, to within 0.01%. Crosses mark allocations that would leave the rest of the world with negative rights. The curve leaves the panel where the equivalent allocation turns negative, that is, where the country prefers the

Table 1: Allocation at which each country is indifferent when the autarky price equals the benchmark ($\pi_i = 1$)

	Emissions p.c. in 2025 (tCO ₂)	Emissions p.c. in 2025 over world average	Predicted $\hat{\rho}_1$	Simulated ρ_1
United States	17.65	3.74	4.25	4.26
Russia	9.92	2.10	2.80	2.80
China	7.95	1.68	0.97	0.95
EU27	6.51	1.38	0.38	0.37
Turkey	4.86	1.03	1.62	1.62
India	1.91	0.40	0.92	0.91
Nigeria	0.59	0.12	0.18	0.18
DR Congo	0.05	0.01	0.03	0.03

Note: $\hat{\rho}_1$ is equation (8) evaluated on the uniform-price run: the country’s emissions over its population share of world emissions, each year weighted by the discounted benchmark price over 2025–2100. ρ_1 is the intercept of the simulated indifference curve in Figure 1, solved on consumption.

5 Equivalent rights of prominent differentiation proposals

5.1 Proposals

We feed three proposals into the model as country-level price schedules.

Wolfram et al. (2025) imagine a coalition of 47 countries⁴ including China, the EU, the UK, India and Brazil, which together emit 48% of world CO₂ in 2030. Members apply \$75, \$50 and \$25/t in high-, upper-middle- and lower-income countries, the tiers of the IMF’s international carbon price floor (Parry et al. 2021). The proposal targets four emissions-intensive industries, whereas NICE2020 has no sectoral detail. We apply the floors economy-wide, so our scenario is more ambitious than the one the authors model.

Banerjee et al. (2025) propose \$10, \$30 and \$50/t in low-, lower-middle- and upper-middle-income countries, in exchange for the compensation of climate damages, with revenues retained domestically. High-income countries fund the compensation and are expected to price carbon heavily, but the proposal gives no

incidence of prices and transfers.

⁴The members are Algeria, Australia, Brazil, Cameroon, Canada, China, Egypt, the European Union, Ghana, Iceland, India, Indonesia, Kenya, Liechtenstein, Mozambique, Norway, Switzerland, Thailand, Togo, the United Kingdom, Uganda and Zambia.

figure. Rather than leaving them unpriced, we price them at \$75/t, the top tier of Wolfram et al. (2025).

Equal Right (2023) proposes a system to phase out fossil fuel extraction that can be modeled as a schedule of differentiated prices. Each country sells its extraction licences at a price graduated by income group (\$240, \$120, \$60 and \$30/t) and discounted by up to 60% for climate vulnerability, which gives floors ranging from \$12 to \$240/t in 2025 (\$144/t on average), rising by about 16% a year to stay within 1.6°C. The revenues are pooled in a global fund that pays climate grants and a universal dividend. We use its schedule of price floors, which is far more dispersed than the other two, together with its global fund: the revenue of every country is pooled and returned as an equal per capita dividend to the world population. Equal Right is therefore the one schedule of the three whose explicit transfers we model, and the only one whose revenue leaves the country that raises it. The report sets a charge for 173 of the 179 economies of the model; we price the remaining six at the median charge of their World Bank income group, so that the schedule covers the world and its club is every country.

The first two proposals do not specify how prices evolve over time. Parry et al. (2021) set their floors for 2030, Wolfram et al. (2025) simulate a single illustrative year and suggest “sunset provisions that gradually raise lower-tier prices so countries ‘graduate’ into a single uniform price”, and Banerjee et al. (2025) raise a country’s price only when it moves to a higher income group. For each proposal, we hold the announced prices from 2025 through 2030 and raise them by 5% a year thereafter, capped at the backstop price. While Wolfram et al. (2025) and Banerjee et al. (2025) include unquantified provisions to make prices converge, we do not model them to properly compare differentiated prices with a cap-and-trade. In the Online Appendix, we also run Equal Right’s original escalation path, which reaches the backstop in most countries by 2050 and 1.48°C of warming in 2100. The proposals yield a 2100 warming of 2.16°C (Wolfram et al. 2025), 1.80°C (Banerjee et al. 2025) and 1.65°C (Equal Right).

5.2 Design

We compare each differentiation proposal with a cap-and-trade. Its members are countries the proposal prices, and they face a uniform price p_t^* , calibrated so that their joint emissions equal those under autarky. This price is \$49.2/t (Wolfram et al. 2025), \$55.8/t (Banerjee et al. 2025) and \$139.4/t (Equal Right) in 2030. It then grows by 3.9%, 4.6% and 3.8% a year respectively until 2050, less than the 5% of the schedules, because baseline emissions shift towards the members with the lowest prices, which weigh more and more in the club’s abatement (and, for Equal Right, because its highest prices reach the backstop from the 2040s). Non-members are unpriced in both regimes. Member i receives rights per capita $r_{it} = \rho_i \bar{e}_t$, where $\bar{e}_t$ is the club’s emissions per capita under the proposal, so that $\rho_i = 1$

is an equal per capita allocation.

We solve for the equivalent ratios ρ_i^* in each of the two variants of Section 4: on the NPV of each member’s total consumption, with Negishi recycling, and on the NPV of its EDE consumption, with an equal per capita dividend within countries (Online Appendix D). Proposition 2 implies that the club regime leaves a surplus, which we use in either of two ways:

- *Reduced emissions:* rights are cut (hence the price and abatement costs increased) until every member is as well off as under the proposal. As in Propositions 1 and 2, members are compared at the same damages: while solving for the allocation, we hold each country’s damages at the level implied by the proposal’s own temperature path. This is an accounting device used to isolate the efficiency gain of uniform pricing. In case the price is below the social cost of carbon, cutting rights actually improves welfare, so accounting for avoided damages would tighten the cap until the price reaches a point above the social cost of carbon. Reduced emissions would then measure the ambition of a proposal rather than the instrument used to deliver it (Online Appendix B). Once rights cuts are computed, we reintegrate avoided damages into the welfare and consumption gains we report.
- *Increased consumption:* joint emissions are kept at their level under the proposal in every year, and the surplus rights are shared so as to maximise the smallest relative gain of any member, which equalises the gains of all members.

We solve each problem jointly for all members. At each iteration, the allocation sets the cap, the club price is recalibrated to deliver it, and each member’s ratio is updated until convergence. In the Online Appendix, we also report an *isolated* solve, which finds each country’s ratio in turn while every other member pays the same uniform price and is granted its own emissions under the proposal as rights. It is the partial-equilibrium analogue of Proposition 1: the cap and the uniform price are held at their levels under the proposal. It therefore misses a general-equilibrium effect: applied together, the isolated ratios add up to a tighter cap, which raises the uniform price, and hence every member’s abatement costs and the value of the rights it buys or sells. With the isolated solve, the surplus is taken either as reduced emissions, by applying the isolated ratios together, or as increased consumption, by scaling the isolated rights up by a common factor each year until they add up to the proposal’s emissions. The Online Appendix also reports two alternative ways of sharing the surplus rights (using either the *joint* or *isolated* solve): in proportion to population times marginal utility, and by uniform scaling.

5.3 Results

Table 2 reports, for ten major economies, the allocation equivalent to each proposal, solved on consumption, together with the outcome of the two ways of using the surplus.

Table 2: Equivalent rights of three differentiated-price schedules, and the dividend of a uniform price

Country	Wolfram et al.		Banerjee, Duflo & Greenstone			Equal Right, 5%/yr	
	p_{2030}	ρ	p_{2030}	$\hat{\rho}$	ρ	p_{2030}	ρ
United States	0	–	75	3.08	2.99	240	–0.34
Russia	0	–	75	1.99	1.90	240	0.00
European Union	75	0.45	75	0.42	0.40	232	0.51
Turkey	0	–	50	1.50	1.50	108	0.99
Indonesia	50	0.96	50	1.05	1.05	96	0.92
China	50	1.06	50	1.04	1.00	108	1.10
Brazil	50	0.46	50	0.48	0.48	108	0.78
India	25	1.12	30	1.16	1.06	42	1.21
Nigeria	0	–	30	0.24	0.22	42	0.77
DR Congo	0	–	10	0.09	0.06	15	0.68
<i>Reduced emissions: every member as well off as under the proposal</i>							
World temp. 2100, change (°C)		–0.010			–0.019		–0.033
Emissions change in the coalition (%)		–4.8%			–4.8%		–13.4%
Post-damage consumption gain (%)		0.023%			0.037%		0.088%
<i>Increased consumption: coalition emissions as under the proposal</i>							
World consumption gain (%)		0.019%			0.050%		0.170%

Note: Joint solve. p_{2030} is the price the schedule asks of the country (\$/t); the comparison regime prices every member of the schedule’s own club at one common price instead (\$49.2/t, \$55.8/t and \$139.4/t respectively in 2030, growing by 3.9%, 4.6% and 3.8% a year until 2050), and leaves non-members unpriced, as the schedule does; of the three, only the [Wolfram et al. \(2025\)](#) schedule has non-members, the other two pricing every country. ρ is the country’s allocation as a multiple of the club’s emissions per capita under the schedule, so that $\rho = 1$ is an equal per capita allocation: it holds the country at its total consumption under the schedule, population-weighted over 2025–2100. The allocations of the welfare variant (NPV of equally-distributed-equivalent consumption, equal per capita dividend within countries) are in Online Appendix D. $\hat{\rho}$ is the first-order prediction of equation (8): the country’s emissions under the schedule over its population share of the club’s, each year weighted by the discounted club price. A country that its schedule does not price is outside that club in both regimes ($p = 0$, no allocation). Under the two schedules of [Wolfram et al. \(2025\)](#) and [Banerjee et al. \(2025\)](#) each country keeps its own revenue; under Equal Right it is pooled worldwide and returned as an equal per capita dividend, as that proposal specifies, which is why its equivalent allocations are so much lower for large per capita emitters and can even be negative. The schedules themselves reach 2.16°C, 1.80°C and 1.65°C in 2100. Under *reduced emissions*, rights are cut until every member is as well off as under the schedule, holding each country’s damages at the level implied by the schedule’s temperature path, so that the cut measures the efficiency gain of uniform pricing and not the avoided damages it brings; the temperature row and the *post-damage consumption gain* are nonetheless computed with damages endogenous, which is what the label records: the members are indifferent at the schedule’s damages, and the gain is what the avoided damages of the tighter cap add on top. Under *increased consumption*, the club’s emissions are held at their level under the schedule and the surplus is shared so as to equalise members’ relative gains: every member gains the same, and the world gain is that common gain times the members’ share of world consumption (about one half for the coalition of [Wolfram et al. \(2025\)](#), one for the two others).

The surplus as reduced emissions. The uniform-price equivalent allocation of the Banerjee et al. differentiation reduces global emissions by 4.8% and its 2100 warming by 0.02°C, while preserving pre-damage consumption in every country. Reduced warming makes every country better off than under autarky except for Mongolia, which would actually benefit from extra warming, according to our model: it loses 0.02% of its consumption, which 0.03 more equal per capita shares (0.45% of its allocation) would compensate (Online Appendix B). The corresponding emission cuts are 4.8% for Wolfram et al. (2025) and 13.4% for the Equal Right schedule. In the welfare variant the cut is far larger (27% for Banerjee et al.), but not because uniform pricing is more efficient: with an equal per capita dividend, a higher club price redistributes within each member, which the welfare criterion values in itself (Online Appendix D).

The surplus as increased consumption. The alternative is to keep total emissions at their level under autarky and to share the surplus rights so as to raise the smallest gain, which equalises the gains of all members. Every country then gains 0.050% of the net present value of its consumption vis-à-vis Banerjee et al.’s proposal. In the welfare variant, every member’s welfare rises by at least 0.105%.

The equivalent rights. Under the schedule of Banerjee et al. (2025), the equivalent rights range from 2.99 equal per capita shares for the United States to 0.06 for the DRC.

The first-order prediction $\hat{\rho}_i$ of equation (8), shown for the Banerjee et al. schedule, reproduces the solved allocations closely. As Proposition 1 implies, it errs on the generous side for most countries, and by most for those priced furthest below the uniform price, whose second-order term is largest: 1.16 against 1.06 for India, 0.09 against 0.06 for the DRC.

The equivalent rights track the two determinants of Proposition 1: a country’s emissions, and the gap between the price it is asked and the club price of \$56/t. The United States and Russia, priced at \$75/t and emitting well above the world average, need 2.99 and 1.90 shares; China, priced at \$50/t, needs 1.00; India, at \$30/t, 1.06; Nigeria, at the same price but emitting a tenth as much per capita, 0.22. The schedule of Wolfram et al. (2025) gives a similar picture among its 47 members. The Equal Right schedule, whose revenue does not stay at home, is a different matter.

A schedule with a global dividend. Under Equal Right the revenue of every country is pooled and paid back as an equal per capita dividend, so the equivalent allocation has to replace what the schedule charges a country *and* what the fund pays it. To first order, a member is indifferent when $p_t^* R_{it} = p_t^* E_{it}^A + D_t N_{it}/N_t - p_{it} E_{it}^A$, with $D_t = \sum_j p_{jt} E_{jt}^A$ the world revenue and $p_{it} E_{it}^A$

foregone revenue from domestic pricing. Denoting by $\bar{p}_t \equiv D_t/E_t$ the average price paid under autarky, the dividend is therefore worth not one equal per capita share but $\bar{p}/p^*$ of one, about 0.75 under this schedule. Indeed, the members charged \$240/t abate almost everything, so the tonnes that survive sit in the \$15–42/t tiers, and the emission-weighted mean price falls far below the uniform price that delivers the same abatement. A member that emits almost nothing therefore ends near $\bar{p}/p^*$ rather than at one: this explains the equivalent right ratio below unity for a country such as the DRC. For eleven countries, the equivalent ratio is negative, e.g. -0.34 for the United States (Russia’s is zero to two decimals), meaning that these countries would have a net duty to remove carbon rather than a net right to emit, in line with the fairness principle accounting for historical responsibility (Fanning & Hickel 2023).

6 Conclusion

Differentiating carbon prices by income is increasingly proposed as a way to reconcile carbon pricing with equity. Yet a differentiated schedule is equivalent, to first order, to a uniform price with tradable rights that grandfather each country on the emissions the schedule itself induces. These are implicit transfers of the same first-order size, only paid in (excess or forgone) abatement rather than in money. Moreover, since abatement costs are convex, the explicit version leaves every country better off.

Replacing the differentiated prices of Wolfram et al. (2025) or Banerjee et al. (2025) with a uniform price at equivalent rights would cut the coalition’s emissions by about 5% while leaving every member as well off as under autarky, and by 13% under the more dispersed schedule of Equal Right (2023). Alternatively, at unchanged emissions, it would raise every member’s consumption.

The equivalence runs only one way. Every schedule of prices has an equivalent allocation of rights, but no schedule of non-negative prices can deliver to a low emitter what a generous allocation would: even a zero price leaves Nigeria below 30% of an equal per capita share, and the DRC below 6%. Allocations based on equal per capita rights, historical responsibility or capacity to pay are therefore out of reach of price differentiation.

Two implications follow for negotiations. First, the equivalence formula provides a near linear correspondence between a price and emission rights, allowing parties to shift the discussion from one notion to the other as needed. Second, the value of national sovereignty invoked to dismiss an international cap-and-trade should be balanced with the efficiency gains that an international commitment would enable. Furthermore, survey evidence shows that 85% of worldwide respondents state that climate policy at the *global* level is needed, at odd with the supposed taste for national sovereignty (Fabre et al. 2025). Finally, robust ma-

majority acceptance of an international cap-and-trade funding equal per capita cash transfers indicates that most people are ready to accept large North–South transfers that differentiated autarky prices cannot deliver.

The analysis has limitations. NICE2020 has capital market, so it cannot capture the diverging capital return motive for differentiation. It has no sectoral detail, which forced us to apply sectoral proposals economy-wide and ignores the country-specific costs of sectoral reallocation. Extending the equivalence to settings with trade and different sectors, where a uniform price must coexist with border adjustments and territorial emissions can be distinguished from carbon footprints, is a natural next step to better inform climate negotiators.

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Declaration of competing interest

The authors declare no competing interests.

Data availability

The code (Julia, built on the Mimi implementation of NICE2020) and all inputs needed to reproduce the results are publicly available on github.com/bixiou/NICE2020.

Declaration of generative AI use

During the preparation of this work the authors used Claude Code (Opus 5) to assist with code and editing. The authors reviewed and edited the content and take full responsibility for it.

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Online Appendix

A Model details

Configuration. In the consumption variant, all runs use NICE2020 with revenue recycling within each country in proportion to c^{η} , computed once from a uniform-price reference run and held fixed across runs; in the welfare variant, revenue is returned to each country's deciles in equal per capita amounts (under the Equal Right schedule the revenue is first pooled worldwide and paid out as an equal per capita dividend, as the proposal specifies, and the variant's rule then shares each country's dividend across its deciles). Both variants use the default income elasticity of emissions across deciles, emissions measured on a consumption basis (carbon footprints), and international transfers entering national income, hence investment. The same configuration is used in the differentiated and in the uniform-price regime, so that the two regimes differ only in prices and rights.

Rest-of-world price in the autarky scenario. When country i applies $\pi_i p_t^*$, the rest of the world applies a price $p_{-i,t}$ such that global emissions follow the path of the uniform-price run. Since every country abates at the rate $\mu = (p/p_t^{\text{back}})^{1/(\theta_2-1)}$, with $\theta_2 = 2.6$, the rest of the world's emissions are $B_{-i,t}(1 - \mu_{-i,t})$, where $B_{-i,t}$ is its baseline. We therefore update $p_{-i,t}$ by Newton steps in the abatement rate, $\mu_{-i,t} \leftarrow 1 - (E_t - E_{it})/B_{-i,t}$, re-evaluating $B_{-i,t}$ in each run, until world emissions are within 0.01% of the target in every year.

Rights of the rest of the world in the uniform-price scenario. Country i receives ρ_i times its population share of global emissions in each year. The remainder is distributed to other countries in proportion to their emissions, which keeps total rights equal to the cap for any ρ_i . When country i 's rights exceed the cap, the remainder is negative and the rest of the world pays for its emissions.

Abatement beyond 100%. Standard implementations constrain the abatement rate to $\mu \in [0, 1]$. Under high autarky prices, country i 's price exceeds the backstop price and this constraint binds. In the autarky scenario we extrapolate the cost function beyond $\mu = 1$, which represents carbon dioxide removal at rising marginal cost. The proposals of Section 5 are instead capped at the backstop price.

Viability flags. We flag grid points at which (i) the rest-of-world price has to be negative in some year to keep the global path, and (ii) country i 's rights exceed

global emissions in some year, leaving the rest of the world with negative rights. Only the second occurs, for China and India at allocations of six equal per capita shares or more (the crosses in Figure 1). The first never does: the lowest rest-of-world price over the whole grid (after 2025, when the benchmark itself is still zero) is \$2.0/t, for China at $\pi_i = 4.75$ in 2026, in both variants.

Table A1: Autarky scenario for the United States ($\pi_i = 2$)

	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	72.8	155.2	182.0	213.3	249.9	292.9	343.3	402.4
p_i (USD/tCO ₂)	145.7	310.5	363.9	426.5	499.9	585.9	686.7	804.8
p_{-i} (USD/tCO ₂)	61.3	127.5	146.7	173.9	208.3	250.5	300.4	359.4
E_i (GtCO ₂)	3.59	1.81	1.01	0.45	-0.09	-0.54	-0.89	-1.12

Table A2: Uniform-price scenario for the Democratic Republic of Congo ($\rho_i = 0.5$)

	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	72.8	155.2	182.0	213.3	249.9	292.9	343.3	402.4
R_i (GtCO ₂)	0.200	0.161	0.145	0.140	0.130	0.111	0.080	0.040
E_i (GtCO ₂)	0.005	0.006	0.009	0.012	0.013	0.012	0.009	0.005
Transfer (bn USD)	14.2	24.0	24.7	27.4	29.2	28.8	24.1	14.3

Note: R_i and E_i are the country's rights and emissions (GtCO₂), and the transfer is the value of its net sales of rights, $(R_i - E_i) p^*$.

B Solves, sharing rules and numerical accuracy

The two solves. The *joint* solve of the main text finds all ρ_i at once. At each iteration the allocation sets the cap, the club price is recalibrated to deliver it, and each member's ratio is moved by a secant step on its gap to the target. For the reduced-emissions variant, an outer loop then moves the level of the allocation until the members' common gain is zero, so that every member ends exactly at its level under the schedule. The *isolated* solve instead finds each country's ratio in turn, with every other member granted its own emissions under the schedule and the club price held at the value that reproduces the schedule's emissions. It is the partial-equilibrium counterpart of Proposition 1: it ignores that the other members also change their allocation, and that the club price adjusts to the resulting cap.

The two solves nearly agree on the aggregate. Taking the equivalent allocation at face value cuts the Wolfram et al. club's emissions by 4.9% (isolated) against

4.8% (joint), and the Banerjee et al. club's by 5.0% against 4.8%. They differ on the distribution, and the difference is systematic. In the isolated solve every other member also pays the uniform price and holds its emissions under the schedule as rights, so each country is stripped of its own efficiency gain only, at an unchanged cap and price, as in Proposition 1. Applied together, the isolated allocations add up to a tighter cap, which raises the uniform price: this raises every member's abatement costs and revalues the rights it trades, to the benefit of net sellers and the detriment of net buyers. The isolated allocations then leave 26 of 47 Wolfram et al. members and 44 of 179 Banerjee et al. members below their level under the schedule, while the joint allocation leaves none, apart from the effect of avoided damages discussed below. Table A3 reports the isolated allocations in the format of Table 2.

Holding damages fixed. In the reduced-emissions variant, the cap is tighter than the schedule's, so warming is lower and every country's damages fall. Propositions 1 and 2 compare the two regimes at the same damages, and so does our solve: when searching for the allocation that leaves each member as well off as under the schedule, we hold each country's damages at the level implied by the schedule's own temperature path. Emissions still fall, and the avoided damages are real; they are simply not allowed to pay for the cut that the solve is measuring. The device is the numerical counterpart of the statement, in Proposition 2, that the cap can be tightened by $\sum_i N_i b_i / p^*$ tonnes with no country worse off: that statement too changes emissions while treating damages as given, and is to be read as a conversion of the surplus into tonnes rather than as a description of a world.

Letting damages respond instead answers a different question. The solve then keeps tightening the cap for as long as the club's avoided damages exceed its marginal abatement cost, that is, until the club price reaches the club's own social cost of carbon. Its answer is the club's optimal cap, which depends on the ambition of climate policy rather than on the instrument that delivers it; in a preliminary solve for Banerjee et al. (2025), whose club is the whole world, the cap had fallen by 21% and every member was still better off when the solve was stopped. We therefore do not report it. The avoided damages that the tighter cap does deliver are reported separately, as the fall in 2100 warming and as the world welfare gain of the variant, both computed with damages endogenous.

Alternative sharing rules. The increased-consumption variant of the main text shares the surplus so as to maximise the smallest relative gain, which equalises the gains of all members. Two other rules are reported in Table A4. Sharing the surplus in proportion to population times marginal utility directs it to the poorest members: it yields by far the largest inequality-averse gain (a world welfare gain

of 0.232% for Banerjee et al., against 0.053% under the maximin rule) but, unlike the maximin rule, it leaves 11 of 179 members below their consumption under the schedule. Uniform scaling of the whole allocation is dominated: the extra rights go to the largest holders, and the looser cap lowers the club price, which devalues the rights of small, generously endowed net sellers. It leaves no member below on consumption, but its inequality-averse gain is the smallest of the three rules (0.040% for Banerjee et al.), and 9 of 47 Wolfram et al. members and 64 of 179 Banerjee et al. members end below their welfare under the schedule.

Who loses from the avoided damages. The equivalence is solved with damages held at the schedule’s temperature path, and every member is then within 0.001% of indifference. The gains reported in Table 2 are computed with damages endogenous, so they also carry the avoided damages of the tighter cap, which are shared in proportion to each country’s marginal damage $\beta_1 + 2\beta_2 T$. In the Kalkuhl–Wenz specification of NICE2020, 13 countries have $\beta_1 < 0$ and a local anomaly below $-\beta_1/2\beta_2$, so their marginal damage is negative: they gain from warming, and less of it makes them worse off. They are the only members that end below their level under the schedule — Mongolia under Banerjee et al. (2025), and Finland, Iceland, Mongolia and Russia under Equal Right at 5%/yr, ordered by $|\beta_1|$ as the mechanism implies, and none under Wolfram et al. (2025), whose cap moves warming least. The loss is a mirror image of the gain: Nigeria and India, whose marginal damage is largest, gain 0.10% and 0.07% of their consumption from the same avoided warming, against Mongolia’s -0.02% ; compensating Mongolia would take 0.03 more equal per capita shares (0.45% of its allocation). This is a property of the damage function, not of the equivalence: at the damages of the schedule, on which the equivalence is defined, no member loses.

Numerical accuracy. The club price is calibrated so that members’ emissions match the cap within 0.01% of the club’s peak emissions in every year. The joint solves stop when no member’s gap to its target exceeds 0.002% of its level under the schedule, and the reduced-emissions outer loop when the common gain is below 0.001%. A member is counted as losing when it falls short by more than 0.005%, four times the tolerance. Prices apply from 2025, and the net present values are taken over 2025–2100, so that the whole period over which the two regimes differ enters the criterion.

Table A3: Allocations equivalent to the three schedules — isolated solve

Country	Wolfram et al.		Banerjee, Duflo & Greenstone			Equal Right, 5%/yr	
	p_{2030}	ρ	p_{2030}	$\hat{\rho}$	ρ	p_{2030}	ρ
United States	0	–	75	3.08	2.88	240	–1.16
Russia	0	–	75	1.99	1.81	240	–0.48
European Union	75	0.42	75	0.42	0.41	232	0.45
Turkey	0	–	50	1.50	1.49	108	0.95
Indonesia	50	0.95	50	1.05	1.05	96	0.96
China	50	1.03	50	1.04	1.01	108	1.04
Brazil	50	0.46	50	0.48	0.48	108	0.86
India	25	1.18	30	1.16	1.08	42	1.35
Nigeria	0	–	30	0.24	0.23	42	0.88
DR Congo	0	–	10	0.09	0.06	15	0.80
<i>Reduced emissions: every member as well off as under the proposal</i>							
World temp. 2100, change (°C)		–0.010			–0.020		–0.041
Emissions change in the coalition (%)		–4.9%			–5.0%		–16.5%
Post-damage consumption gain (%)		0.024%			0.037%		0.077%
<i>Increased consumption: coalition emissions as under the proposal</i>							
World consumption gain (%)		0.019%			0.050%		0.171%

Note: As Table 2, with each country's allocation solved on its own while every other member holds its emissions under the schedule and the club price is held at the value that reproduces the schedule's emissions.

C The Equal Right schedule on its own escalation path

The Equal Right schedule is designed to rise by about 16% a year, far faster than the 5% we assume for the other two. Table A6 reports it on both paths. On its own path, the schedule pushes most members to full abatement by 2050 and reaches 1.48°C of warming; its club price is far above the other schedules', and its allocations are compressed towards an equal per capita share, the dividend of the global fund weighing more heavily the higher the price. The dividend of uniform pricing is larger still: replacing the schedule with the club price and the equivalent allocation cuts the club's emissions by 14.4%, against 13.4% on the 5%-a-year path. Its effect on 2100 warming is smaller (–0.010°C against –0.033°C), because a schedule that has already driven emissions to zero by mid-century leaves little room for a further cut in cumulative terms.

Table A4: Club-price rights ratios ρ_i equivalent to the Wolfram et al. and Banerjee, Duflo & Greenstone proposals

Country	Wolfram et al.			Banerjee, Duflo & Greenstone		
	p_i	ρ^{isolated}	ρ^{joint}	p_i	ρ^{isolated}	ρ^{joint}
United States	0	–	–	75	2.88	2.99
Russia	0	–	–	75	1.81	1.90
European Union	75	0.42	0.45	75	0.41	0.40
Turkey	0	–	–	50	1.49	1.50
Indonesia	50	0.95	0.96	50	1.05	1.05
China	50	1.03	1.06	50	1.01	1.00
Brazil	50	0.46	0.46	50	0.48	0.48
India	25	1.18	1.12	30	1.08	1.06
Nigeria	0	–	–	30	0.23	0.22
DR Congo	0	–	–	10	0.06	0.06
<i>Reduced emissions (variant 1)</i>						
Emissions change in the coalition (%)		–4.9%	–4.8%		–5.0%	–4.8%
World temp. 2100, change (°C)		–0.010	–0.010		–0.020	–0.019
World welfare gain (%)		0.036%	0.027%		0.080%	0.072%
World consumption gain (%)		0.024%	0.023%		0.037%	0.037%
<i>Increased consumption (variant 2)</i>						
World welfare gain (%)		0.008%	0.005%		0.046%	0.053%
World consumption gain (%)		0.019%	0.019%		0.050%	0.050%
<i>Surplus shared by marginal utility (variant 3)</i>						
World welfare gain (%)		0.045%	0.037%		0.243%	0.232%
World consumption gain (%)		0.021%	0.020%		0.054%	0.054%
<i>Surplus shared by uniform scaling (variant 4)</i>						
World welfare gain (%)		=(2)	–0.000%		=(2)	0.040%
World consumption gain (%)		=(2)	0.018%		=(2)	0.049%

Note: p_i is the carbon price the proposal asks of country i in 2030; the comparison regime instead prices every member of the proposal's own club at one common price (\$49.2/t for Wolfram et al. and \$55.8/t for Banerjee, Duflo & Greenstone in 2030) and leaves non-members unpriced, as the proposal does. ρ_i is the country's allocation as a multiple of an equal-per-capita share of the club's emissions under the proposal, taken at face value in variant 1; ρ^{isolated} solves each country on its own, ρ^{joint} all of them jointly. The European Union figure aggregates its members' rights, and a country the proposal does not price is outside the club in both regimes, shown at $p_i = 0$ with no equivalent allocation ('–'). Variant 1 (reduced emissions) lets the equivalent allocation set the cap: in the joint solve every member is held exactly at its level under the proposal; in the isolated solve each country's ratio is solved with all others grandfathered, and the ratios are then applied together. Variants 2 to 4 (increased consumption) keep club emissions at the proposal's and share the surplus: equalising the relative gains of all members (joint variant 2), by scaling the isolated allocation up uniformly (isolated variant 2, also variant 4), and in proportion to population times marginal utility (variant 3). The temperature row is the change from the proposal's own 2100 warming, which is 2.16°C for Wolfram et al. and 1.80°C for Banerjee, Duflo & Greenstone. World welfare is the NPV of the inequality-averse global EDE consumption ($\eta = 1.5$); world consumption is mean consumption per capita, which weights every person's consumption equally.

Table A5: Club members left below their welfare under the proposal, by variant

Variant	Wolfram et al. (of 47)		Banerjee, Duflo & Greenstone (of 179)	
	isolated	joint	isolated	joint
<i>Members losing on equally-distributed-equivalent consumption</i>				
Variant 1	6	1	55	38
Variant 2	7	2	81	42
Variant 3	2	2	34	31
Variant 4	7	9	81	64
<i>Members losing on mean consumption per capita</i>				
Variant 1	26	0	44	1
Variant 2	5	0	49	0
Variant 3	0	2	0	11
Variant 4	5	0	49	0

Note: a member counts as losing when its NPV of consumption over 2025–2100 falls short of what it obtains under the proposal itself. Variant 1 makes every member indifferent by construction, so any count there is numerical noise. Variants 2 to 4 return the same surplus of rights under different sharing rules: a split chosen to minimise the largest relative shortfall (2), one proportional to population times marginal utility (3), and uniform scaling (4).

D The welfare variant

In the welfare variant, a country’s situation is measured by the net present value over 2025–2100 of its population-weighted equally-distributed-equivalent consumption, and the carbon revenue a country keeps (or, under Equal Right, the dividend it receives from the global fund) is returned to its deciles as an equal per capita dividend, in the differentiated and in the uniform regime alike. Everything else — the benchmark price, the schedules and the solves — is as in the consumption variant. Figure A1 is the analogue of Figure 1, and Table A7 reports the joint solves of Table 2 under both variants.

The two variants differ through a within-country channel. In NICE2020, carbon costs fall on deciles in proportion to income raised to an elasticity below one, which is regressive; an equal per capita dividend more than offsets it, so that a carbon price, net of the dividend it funds, redistributes from richer to poorer deciles, which an inequality-averse criterion values. In the consumption variant, the revenue is returned in proportion to c^η and the criterion ignores the distribution across deciles, so that channel is shut down.

Equivalent rights by country. At $\pi_i = 1$ the two variants give nearly the same allocations: ρ_1 is 4.26 for the United States, 0.96 for China, 0.38 for the EU27, 0.94 for India and 0.18 for Nigeria, within 0.03 of the consumption variant (Table 1).

Table A6: The Equal Right schedule on its own escalation path and at 5% a year

Country	Equal Right		Equal Right, 5%/yr	
	p_{2030}	ρ	p_{2030}	ρ
Russia	513	0.18	240	0.00
United States	513	-0.09	240	-0.34
European Union	496	0.43	232	0.51
China	231	1.13	108	1.10
Turkey	231	0.96	108	0.99
Brazil	231	0.77	108	0.78
Indonesia	205	0.87	96	0.92
India	90	1.10	42	1.21
Nigeria	90	0.77	42	0.77
DR Congo	32	0.67	15	0.68
<i>Reduced emissions: every member as well off as under the proposal</i>				
World temp. 2100, change (°C)		-0.010		-0.033
Emissions change in the coalition (%)		-14.4%		-13.4%
Post-damage consumption gain (%)		0.034%		0.088%
<i>Increased consumption: coalition emissions as under the proposal</i>				
World consumption gain (%)		0.093%		0.170%

Note: As Table 2, for the Equal Right schedule of price floors grown at its own rate (about 16% a year) and at the 5% a year assumed for the other schedules. Both are run with the proposal's global fund: the revenue of every country is pooled and returned as an equal per capita dividend to the world population.

Away from $\pi_i = 1$ they differ. Because a domestic price also funds a progressive dividend, a country's welfare under autarky hardly falls with its own price at low π_i , and the equivalent allocation is highest around $\pi_i = 0.5$ –1 rather than at a zero price: 4.52 at $\pi_i = 0.5$ against 3.58 at $\pi_i = 0$ for the United States, 0.19 at most against 0.14 at $\pi_i = 0$ for Nigeria, and 0.04 at most for the DRC. The most that any non-negative price can offer a low emitter is therefore even smaller than in the consumption variant, which strengthens the conclusion of Section 4. The colour scale is wider (± 35 against ± 15.1): an allocation's transfer is shared equally across deciles, and it raises equally-distributed-equivalent consumption, which is below mean consumption, by a larger percentage.

Equivalent rights of the proposals. The surplus left by uniform pricing is far larger in the welfare variant (Table A7). Converted into lower emissions, it reaches 34.5% of the coalition's emissions under Wolfram et al. (2025), 27.2% under Banerjee et al. (2025) and 29.6% under Equal Right, against 4.8%, 4.8% and 13.4% in the consumption variant. The equivalent allocations, which add up to these tighter caps, are accordingly lower, and the difference falls mostly on middle- and low-income members (India 0.62 against 1.06 under Banerjee et al., China 0.82 against 1.00), while high emitters keep similar allocations (the United States 2.83 against 2.99). This is not a larger efficiency gain from uniform pricing. In the welfare variant a tighter cap raises members' welfare by itself, even at the damages of the schedule: at a higher club price each member collects more revenue, whose equal per capita return redistributes within the country by more than the extra abatement costs. The members' common gain first rises as the cap is tightened (for Banerjee et al., from 0.068% to 0.072% of their welfare when the allocation is cut by a further tenth) and only falls back to zero once the cap is 27% below the proposal's emissions. The reduced-emissions cut of the welfare variant thus mixes the efficiency gain of uniform pricing with the within-country redistributive value of carbon revenue; the consumption variant isolates the former. Holding emissions at the proposals' level instead, the surplus shared by the maximin rule raises every member's welfare by at least 0.078% (Wolfram et al.), 0.105% (Banerjee et al.) and 0.371% (Equal Right).

Table A7: Equivalent rights of the three schedules in the two variants (joint solve)

Country	Wolfram et al.			Banerjee, Duflo & Greenstone			Equal Right, 5%/yr		
	p_{2030}	ρ^{cons}	ρ^{welf}	p_{2030}	ρ^{cons}	ρ^{welf}	p_{2030}	ρ^{cons}	ρ^{welf}
United States	0	–	–	75	2.99	2.83	240	–0.34	0.39
European Union	75	0.45	0.47	75	0.40	0.45	232	0.51	0.54
Russia	0	–	–	75	1.90	1.79	240	0.00	0.43
China	50	1.06	0.81	50	1.00	0.82	108	1.10	0.85
Turkey	0	–	–	50	1.50	1.13	108	0.99	0.81
Brazil	50	0.46	0.27	50	0.48	0.29	108	0.78	0.58
Indonesia	50	0.96	0.72	50	1.05	0.77	96	0.92	0.73
India	25	1.12	0.59	30	1.06	0.62	42	1.21	0.81
Nigeria	0	–	–	30	0.22	0.12	42	0.77	0.59
DR Congo	0	–	–	10	0.06	0.03	15	0.68	0.55
<i>Reduced emissions: every member as well off as under the proposal</i>									
World temp. 2100, change (°C)		–0.010	–0.068		–0.019	–0.108		–0.033	–0.073
Emissions change in the coalition (%)		–4.8%	–34.5%		–4.8%	–27.2%		–13.4%	–29.6%
Post-damage consumption gain (%)		0.023%	–0.012%		0.037%	–0.020%		0.088%	–0.052%
Post-damage welfare gain (%)		0.027%	0.322%		0.072%	0.618%		0.076%	0.389%
<i>Increased consumption: coalition emissions as under the proposal</i>									
World consumption gain (%)		0.019%	0.016%		0.050%	0.045%		0.170%	0.164%
World welfare gain (%)		0.005%	0.025%		0.053%	0.224%		0.064%	0.408%

Note: As Table 2, for the two variants. ρ^{cons} : consumption variant (NPV of total consumption, revenue recycled within countries in proportion to c^{η}), as in Table 2. ρ^{welf} : welfare variant (NPV of population-weighted equally-distributed-equivalent consumption, revenue returned as an equal per capita dividend within countries). Each column is solved on its own criterion, both over 2025–2100; each summary row is computed for both. The consumption gains are computed on world consumption, the welfare gains on the global equally-distributed-equivalent consumption, both with damages endogenous. The schedules reach the same 2100 warming in both variants (2.16°C, 1.80°C and 1.65°C).

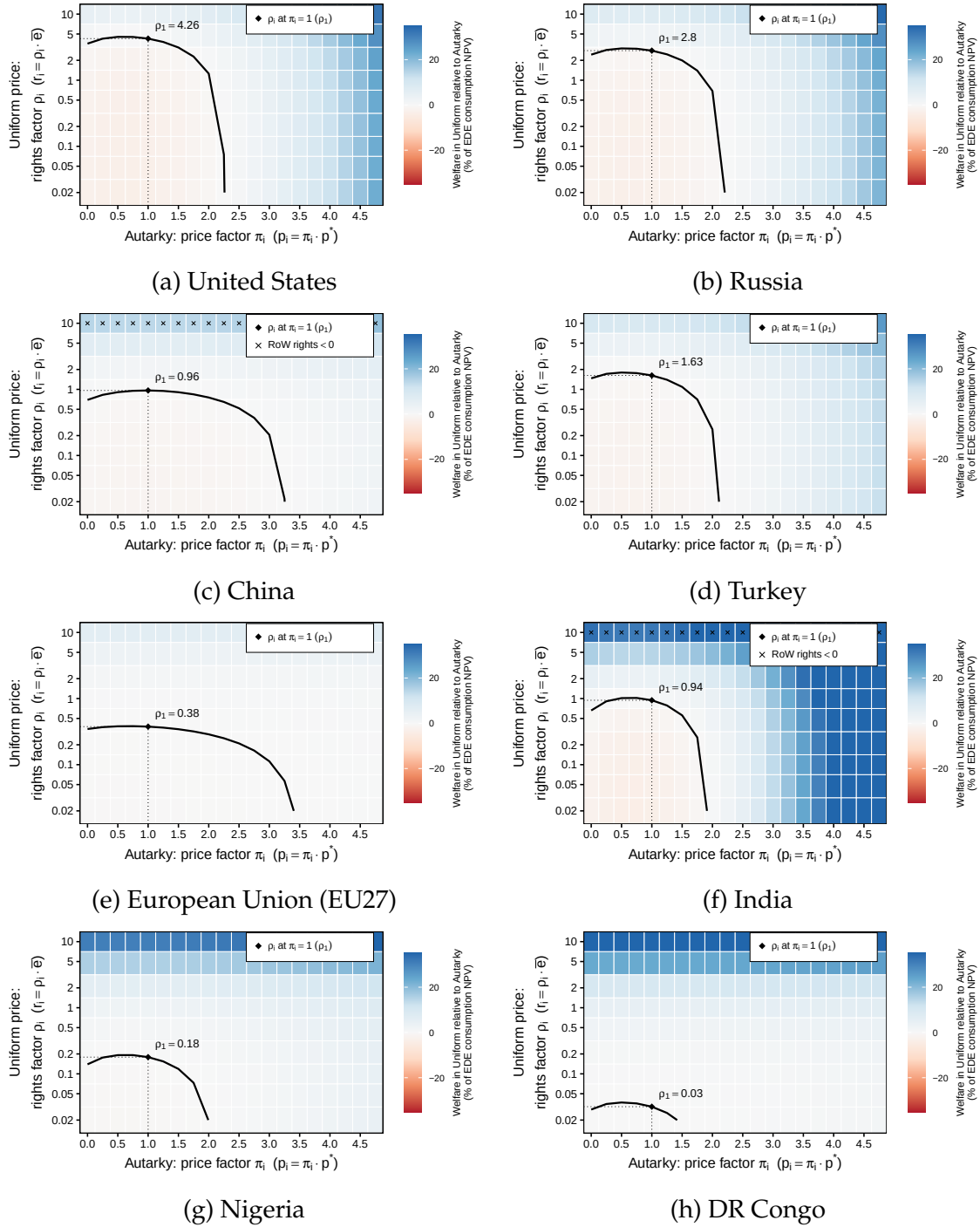


Figure A1: Welfare gain of the uniform-price regime over autarky, welfare variant

Note: As Figure 1, for the welfare variant: colours give the difference in the NPV of population-weighted equally-distributed-equivalent consumption over 2025–2100, in percent, clipped at ± 35 (the 95th percentile of the absolute difference), and the carbon revenue is returned as an equal per capita dividend within countries in both regimes. The indifference curve is traced from the equivalent allocation solved at each π_i of the grid.